Differentiation & Integration

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Derivatives of Polynomials & Tangents

Find the x-coordinates of the stationary points of $y = x^3 + 15/2x^2 + 18x$.

Final Answer: $x = -3$ or $x = -2$. $\frac{dy}{dx} = 3x^2 + 15x + 18 = 3(1)(x+2)(x+3) = 0 \Rightarrow x = -3$ or $x = -2$

2 [Foundational] • Target: Stationary Points & Optimization

Find the x-coordinates of the stationary points of $y = 2x^3 + 12x^2 + 18x$.

Final Answer: $x = -3$ or $x = -1$. $\frac{dy}{dx} = 6x^2 + 24x + 18 = 3(2)(x+3)(x+1) = 0 \Rightarrow x = -3$ or $x = -1$

3 [Foundational] • Target: Indefinite & Definite Integrals

Find the x-coordinates of the stationary points of $y = 2x^3 - 9x^2 - 24x$.

Final Answer: $x = -1$ or $x = 4$. $\frac{dy}{dx} = 6x^2 - 18x - 24 = 3(2)(x-4)(x+1) = 0 \Rightarrow x = -1$ or $x = 4$

4 [Foundational] • Target: Area Under Curves & Exam-Style Modeling

Find the x-coordinates of the stationary points of $y = 3x^3 + 9x^2 - 72x$.

Final Answer: $x = -4$ or $x = 2$. $\frac{dy}{dx} = 9x^2 + 18x - 72 = 3(3)(x+4)(x-2) = 0 \Rightarrow x = -4$ or $x = 2$

5 [Foundational] • Target: Derivatives of Polynomials & Tangents

Find the x-coordinates of the stationary points of $y = 3x^3 - 9/2x^2 - 54x$.

Final Answer: $x = -2$ or $x = 3$. $\frac{dy}{dx} = 9x^2 - 9x - 54 = 3(3)(x-3)(x+2) = 0 \Rightarrow x = -2$ or $x = 3$

6 [Foundational] • Target: Stationary Points & Optimization

Find the x-coordinates of the stationary points of $y = x^3 + 3/2x^2 - 18x$.

Final Answer: $x = -3$ or $x = 2$. $\frac{dy}{dx} = 3x^2 + 3x - 18 = 3(1)(x+3)(x-2) = 0 \Rightarrow x = -3$ or $x = 2$

7 [Foundational] • Target: Indefinite & Definite Integrals

Find the x-coordinates of the stationary points of $y = x^3 - 3/2x^2 - 6x$.

Final Answer: $x = -1$ or $x = 2$. $\frac{dy}{dx} = 3x^2 - 3x - 6 = 3(1)(x+1)(x-2) = 0 \Rightarrow x = -1$ or $x = 2$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

© ClipSAT — clipsat.org | All Rights Reserved

8 [Intermediate] • Target: Area Under Curves & Exam-Style Modeling

Find the x-coordinates of the stationary points of $y = 2x^3 + 12x^2$.

Final Answer: $x = -4$ or $x = 0$. $\frac{dy}{dx} = 6x^2 + 24x = 3(2)(x - 0)(x + 4) = 0 \Rightarrow x = -4$ or $x = 0$

9 [Intermediate] • Target: Derivatives of Polynomials & Tangents

Find the x-coordinates of the stationary points of $y = x^3 - 12x$.

Final Answer: $x = -2$ or $x = 2$. $\frac{dy}{dx} = 3x^2 - 12 = 3(1)(x + 2)(x - 2) = 0 \Rightarrow x = -2$ or $x = 2$

10 [Intermediate] • Target: Stationary Points & Optimization

Find the area enclosed between the curve $y = 4x - 4$, the x-axis, and the lines $x = 1$ and $x = 4$.

Final Answer: 18. Area $= \int_1^4 (4x - 4)dx = [2x^2 - 4x]_1^4 = 18$

11 [Intermediate] • Target: Indefinite & Definite Integrals

Find the area enclosed between the curve $y = 4x - 8$, the x-axis, and the lines $x = 2$ and $x = 4$.

Area $= \int_2^4 (4x - 8)dx = [2x^2 - 8x]_2^4 = 8$

12 [Intermediate] • Target: Area Under Curves & Exam-Style Modeling

Find the area enclosed between the curve $y = 2x^2$, the x-axis, and the lines $x = 0$ and $x = 4$.

Final Answer: 128/3. Area $= \int_0^4 (2x^2)dx = [2/3x^3]_0^4 = 128/3$

13 [Intermediate] • Target: Derivatives of Polynomials & Tangents

Find the area enclosed between the curve $y = x^2$, the x-axis, and the lines $x = 0$ and $x = 4$.

Final Answer: 64/3. Area $= \int_0^4 (x^2)dx = [1/3x^3]_0^4 = 64/3$

14 [Intermediate] • Target: Stationary Points & Optimization

Find the area enclosed between the curve $y = 2x^2$, the x-axis, and the lines $x = 0$ and $x = 3$.

Final Answer: 18. Area $= \int_0^3 (2x^2)dx = [2/3x^3]_0^3 = 18$

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Indefinite & Definite Integrals • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

Find the area enclosed between the curve $y = 3x$, the x-axis, and the lines $x = 0$ and $x = 3$.

Final Answer: 27/2. Area $= \int_0^3 (3x)dx = [3/2x^2]_0^3 = 27/2$

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Area Under Curves & Exam-Style Modeling • Cross-Lesson: 3.01 Graphs & Transformations

Find the area enclosed between the curve $y = 3x$, the x-axis, and the lines $x = 0$ and $x = 5$.

Final Answer: 75/2. Area $= \int_0^5 (3x)dx = [3/2x^2]_0^5 = 75/2$

17 [Challenging] • Target: Derivatives of Polynomials & Tangents • Cross-Lesson: 4.01 Coordinate Geometry

Find the area enclosed between the curve $y = 3x^2$, the x-axis, and the lines $x = 0$ and $x = 5$.

Final Answer: 125. Area $= \int_0^5 (3x^2)dx = [x^3]_0^5 = 125$

18 [Challenging] • Target: Stationary Points & Optimization • Cross-Lesson: 5.01 Trigonometry

Find the area enclosed between the curve $y = 3x^2$, the x-axis, and the lines $x = 0$ and $x = 3$.

Final Answer: 27. Area $= \int_0^3 (3x^2)dx = [x^3]_0^3 = 27$

19 [Challenging] • Target: Indefinite & Definite Integrals • Cross-Lesson: 7.01 Statistics & Probability

Find the area enclosed between the curve $y = x$, the x-axis, and the lines $x = 0$ and $x = 3$.

Final Answer: 9/2. Area $= \int_0^3 (x)dx = [1/2x^2]_0^3 = 9/2$

20 [Challenging] • Target: Area Under Curves & Exam-Style Modeling • Cross-Lesson: 8.01 Functions: Composite & Inverse

Find the area enclosed between the curve $y = 3x^2$, the x-axis, and the lines $x = 0$ and $x = 2$.

Final Answer: 8. Area $= \int_0^2 (3x^2)dx = [x^3]_0^2 = 8$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Derivatives of Polynomials & Tangents

Differentiate $y = 3x^4 - 5x^2 + 7x - 2$ with respect to x.

a. $12x^3 - 10x + 7$ ✓

b. $12x^3 - 10x^2 + 7$

c. $7x^3 - 10x + 7$

d. $12x^4 - 10x^2 + 7$

Apply the power rule: $d/dx[3x^4] = 12x^3$, $d/dx[-5x^2] = -10x$, $d/dx[7x] = 7$, $d/dx[-2] = 0$.

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: Derivatives of Polynomials & Tangents

Find the gradient of the curve $y = x^3 - 4x + 1$ at the point where $x = 2$.

a. **8** ✓

b. 12

c. 4

d. 0

$dy/dx = 3x^2 - 4$. At $x = 2$: $dy/dx = 3(4) - 4 = 12 - 4 = 8$.

23 [Foundational] • Target: Derivatives of Polynomials & Tangents

Find the equation of the tangent to the curve $y = x^2 - 3x + 2$ at the point $(3, 2)$.

a. $y = 3x - 7$ ✓

b. $y = 3x - 2$

c. $y = x - 1$

d. $y = 2x - 4$

$dy/dx = 2x - 3$. At $x = 3$, gradient $m = 2(3) - 3 = 3$. Tangent: $y - 2 = 3(x - 3) \Rightarrow y = 3x - 7$.

24 [Foundational] • Target: Stationary Points & Optimization

Find the stationary points of the curve $y = 2x^3 - 9x^2 + 12x + 1$ and determine their nature.

a. **Local max at** $(1, 6)$**, local min at** $(2, 5)$ ✓

b. Local min at $(1, 6)$, local max at $(2, 5)$

c. Local max at $(2, 5)$, inflection at $(1, 6)$

d. Local min at $(0, 1)$, local max at $(2, 5)$

$dy/dx = 6x^2 - 18x + 12 = 6(x^2 - 3x + 2) = 6(x - 1)(x - 2) = 0 \Rightarrow x = 1, 2$. $d^2y/dx^2 = 12x - 18$. At $x = 1$: $d^2y/dx^2 = -6 < 0$ (max, $y = 6$). At $x = 2$: $d^2y/dx^2 = 6 > 0$ (min, $y = 5$).

25 [Foundational] • Target: Stationary Points & Optimization

A metal wire of length 40 cm is bent into a rectangle of dimensions x by y. What value of x maximizes the enclosed area?

a. $x = 10$ cm ✓

b. $x = 20$ cm

c. $x = 5$ cm

d. $x = 15$ cm

$2x + 2y = 40 \Rightarrow y = 20 - x$. Area $A = x(20 - x) = 20x - x^2$. $dA/dx = 20 - 2x = 0 \Rightarrow x = 10$.

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Stationary Points & Optimization

For what range of values of x is the function $f(x) = x^3 - 6x^2 + 9x - 4$ strictly increasing?

a. $x < 1$ **or** $x > 3$ ✓

b. $1 < x < 3$

c. $x > 2$

d. $x < 2$

$f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3) > 0 \Rightarrow x < 1$ or $x > 3$.

27 [Foundational] • Target: Indefinite & Definite Integrals

Find $\int (6x^2 - 4x + 5)\,dx$.

a. $2x^3 - 2x^2 + 5x + C$ ✓

b. $3x^3 - 2x^2 + 5x + C$

c. $2x^3 - 4x^2 + 5x + C$

d. $12x - 4 + C$

$\int 6x^2 dx = 2x^3$, $\int -4x dx = -2x^2$, $\int 5dx = 5x$. Result: $2x^3 - 2x^2 + 5x + C$.

28 [Foundational] • Target: Indefinite & Definite Integrals

Evaluate the definite integral $\int_1^3 (3x^2 - 2x)\,dx$.

a. **18** ✓

b. 24

c. 16

d. 20

$[x^3 - x^2]_1^3 = (27 - 9) - (1 - 1) = 18 - 0 = 18$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Indefinite & Definite Integrals

A curve passes through $(1, 4)$ and has gradient function $\frac{dy}{dx} = 4x - 1$. Find the equation of the curve.

a. $y = 2x^2 - x + 3$ ✓

b. $y = 2x^2 - x + 4$

c. $y = 4x^2 - x + 1$

d. $y = 2x^2 - x + 1$

$y = \int (4x - 1)dx = 2x^2 - x + C$. At $(1, 4)$: $4 = 2(1) - 1 + C \Rightarrow 4 = 1 + C \Rightarrow C = 3$. Curve: $y = 2x^2 - x + 3$.

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Area Under Curves & Exam-Style Modeling

Find the area enclosed by the curve $y = 4 - x^2$ and the x-axis.

a. $\frac{32}{3}$ ✓

b. $\frac{16}{3}$

c. 8

d. $\frac{64}{3}$

Roots at $x = \pm 2$. Area $= \int_{-2}^{2}(4 - x^2)dx = [4x - x^3/3]_{-2}^{2} = (8 - 8/3) - (-8 + 8/3) = 16/3 - (-16/3) = 32/3$.

31 [Intermediate] • Target: Area Under Curves & Exam-Style Modeling

Find the area of the region bounded by the curve $y = x^2$ and the line $y = 2x$.

a. $\frac{4}{3}$ ✓

b. $\frac{2}{3}$

c. 2

d. $\frac{8}{3}$

Intersections: $x^2 = 2x \Rightarrow x(x - 2) = 0 \Rightarrow x = 0, 2$. Area $= \int_0^2 (2x - x^2)dx = [x^2 - x^3/3]_0^2 = 4 - 8/3 = 4/3$.

32 [Intermediate] • Target: Area Under Curves & Exam-Style Modeling

A particle moves along a straight line such that its velocity at time t is $v(t) = 3t^2 - 12t + 9$. Find the total displacement between $t = 0$ and $t = 3$.

a. **0** ✓

b. 9

c. 18

d. 4.5

Displacement $= \int_0^3 (3t^2 - 12t + 9)dt = [t^3 - 6t^2 + 9t]_0^3 = (27 - 54 + 27) - 0 = 0$.

33 [Intermediate] • Target: Derivatives of Polynomials & Tangents

Find the area enclosed between the curve $y = 3x$, the x-axis, and the lines $x = 0$ and $x = 2$.

a. 11

b. 5

c. 12

d. **6** ✓

Area $= \int_0^2 (3x)dx = [3/2x^2]_0^2 = 6$

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Stationary Points & Optimization

Find the area enclosed between the curve $y = 6x$, the x-axis, and the lines $x = 0$ and $x = 6$.

a. 107

b. 113

c. **108** ✓

d. 216

Area $= \int_0^6 (6x)dx = [3x^2]_0^6 = 108$

35 [Intermediate] • Target: Indefinite & Definite Integrals

Find the area enclosed between the curve $y = 6x$, the x-axis, and the lines $x = 0$ and $x = 4$.

a. 47

b. 53

c. 96

d. **48** ✓

Area $= \int_0^4 (6x)dx = [3x^2]_0^4 = 48$

36 [Intermediate] • Target: Area Under Curves & Exam-Style Modeling

Find the area enclosed between the curve $y = 4x$, the x-axis, and the lines $x = 0$ and $x = 4$.

a. **32** ✓

b. 64

c. 37

d. 31

Area $= \int_0^4 (4x)dx = [2x^2]_0^4 = 32$

37 [Intermediate] • Target: Derivatives of Polynomials & Tangents

Find the area enclosed between the curve $y = 2x$, the x-axis, and the lines $x = 0$ and $x = 4$.

a. 15

b. 21

c. 32

d. **16** ✓

Area $= \int_0^4 (2x)dx = [x^2]_0^4 = 16$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: Stationary Points & Optimization • Cross-Lesson: 5.01 Trigonometry

Find the area enclosed between the curve $y = 2x$, the x-axis, and the lines $x = 0$ and $x = 6$.

a. 72

b. 35

c. 41

d. **36** ✓

Area $= \int_0^6 (2x)dx = [x^2]_0^6 = 36$

39 [Challenging] • Target: Indefinite & Definite Integrals • Cross-Lesson: 7.01 Statistics & Probability

Find the area enclosed between the curve $y = 2x$, the x-axis, and the lines $x = 0$ and $x = 2$.

a. **4** ✓

b. 3

c. 8

d. 9

Area $= \int_0^2 (2x)dx = [x^2]_0^2 = 4$

40 [Challenging] • Target: Area Under Curves & Exam-Style Modeling • Cross-Lesson: 8.01 Functions: Composite & Inverse

Find the area enclosed between the curve $y = 2x$, the x-axis, and the lines $x = 0$ and $x = 3$.

a. 8

b. 18

c. 14

d. **9** ✓

Area $= \int_0^3 (2x)dx = [x^2]_0^3 = 9$

41 [Challenging] • Target: Derivatives of Polynomials & Tangents • Cross-Lesson: 9.01 Circular Measure

Find the x-coordinate of the stationary point of $y = 4x^2 + 10x - 3$.

a. -1/4

b. 5/4

c. **-5/4** ✓

d. -9/4

$\frac{dy}{dx} = 8x + 10 = 0 \Rightarrow x = -5/4$

42 [Challenging] • Target: Stationary Points & Optimization • Cross-Lesson: 1.01 Indices, Surds & Quadratics

Find the x-coordinate of the stationary point of $y = 3(x + 1)^2 - 2$.

a. 1

b. 0

c. **-1** ✓

d. -2

$\frac{dy}{dx} = 6(x + 1) = 0 \Rightarrow x = -1$

© ClipSAT — clipsat.org | All Rights Reserved

43 [Challenging] • Target: Indefinite & Definite Integrals • Cross-Lesson: 10.01 Series & Binomial Theorem

Find the x-coordinate of the stationary point of $y = x^2 + x + 8$.

a. **-1/2 ✓**

b. 1/2

c. -3/2

d. 5/2

$\frac{dy}{dx} = 2x + 1 = 0 \Rightarrow x = -1/2$

44 [Challenging] • Target: Area Under Curves & Exam-Style Modeling • Cross-Lesson: 11.01 Kinematics

Find the x-coordinate of the stationary point of $y = (x - 5)^2 - 5$.

a. -5

b. **5 ✓**

c. 4

d. 6

$\frac{dy}{dx} = 2(x - 5) = 0 \Rightarrow x = 5$

45 [Challenging] • Target: Derivatives of Polynomials & Tangents • Cross-Lesson: 12.01 Forces & Equilibrium

Find the x-coordinate of the stationary point of $y = (x - 4)^2 + 4$.

a. -4

b. **4 ✓**

c. 5

d. 3

$\frac{dy}{dx} = 2(x - 4) = 0 \Rightarrow x = 4$

© ClipSAT — clipsat.org | All Rights Reserved